

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0611 COURSE NAME: MACHINE LEARNING

CO5 – Pseudocode Writing Task (Algorithm Design)

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Question 1: Data Cleaning and Transformation on Time-Series Data

Design a pseudocode algorithm to perform data cleaning and transformation on time-series data. The algorithm handles missing timestamps, interpolates missing values, removes outliers, resamples data at fixed intervals, indexes by time, filters invalid entries, and maintains chronological consistency.

Pseudocode

ALGORITHM Clean_Time_Series_Data

INPUT: Time-series dataset D

OUTPUT: Cleaned and resampled dataset C

BEGIN

1. Load dataset D.
2. Check that the timestamp column exists.
3. Convert timestamp values to a standard date-time format.
4. Remove rows having invalid or unparseable timestamps.
5. Remove duplicate timestamp records where required.
6. Sort D in ascending order according to timestamp.
7. Set timestamp as the time index.
8. Identify numeric measurement columns.
9. For each numeric column:
 - a. Detect missing values.
 - b. Interpolate missing values using time-based interpolation.
 - c. If values remain missing at the beginning or end,

apply an appropriate forward-fill or backward-fill rule.

10. For each numeric column:
 - a. Calculate Q1 and Q3.
 - b. Calculate $IQR = Q3 - Q1$.
 - c. Define lower bound = $Q1 - 1.5 * IQR$.
 - d. Define upper bound = $Q3 + 1.5 * IQR$.
 - e. Identify values outside the bounds as outliers.
 - f. Remove or cap the outliers according to the selected policy.
11. Filter invalid entries such as impossible negative values where the domain does not permit them.
12. Resample the cleaned time series at the required fixed interval such as hourly, daily, or weekly.
13. Apply an appropriate aggregation function such as mean, sum, minimum, or maximum during resampling.
14. Verify that the resulting timestamps are unique and chronological.
15. Check that the expected interval is maintained.
16. Store the cleaned dataset as C.
17. Return C.

END

Efficiency Considerations

Use vectorized operations for missing-value detection, interpolation, filtering, and aggregation. Sort the timestamps once before time-based operations and avoid unnecessary repeated scans of the dataset.

Question 2: Feature Selection

Design a feature-selection algorithm that computes correlation between features, removes redundant or irrelevant attributes, retains significant features, normalizes the data, applies threshold-based filtering, and validates whether dimensionality reduction improves model performance.

Pseudocode

ALGORITHM Feature_Selection

INPUT: Dataset D with features X and target Y

OUTPUT: Selected feature set S

BEGIN

1. Load dataset D.
2. Separate predictor features X from target variable Y.
3. Handle missing or invalid values.
4. Identify numeric and categorical features.
5. Normalize or standardize numeric features.
6. Remove features with zero or near-zero variance.
7. Compute the correlation matrix among numeric features.
8. Set a correlation threshold T, for example 0.80.
9. For each pair of features (F_i, F_j):
 - a. If $\text{ABS}(\text{CORRELATION}(F_i, F_j)) \geq T$:
 - i. Compare their relevance to target Y.
 - ii. Retain the more relevant feature.
 - iii. Mark the redundant feature for removal.
10. For each remaining feature:
 - a. Measure its relationship with target Y using an appropriate statistical test or feature-importance method.
 - b. Remove the feature if its relevance is below the predefined significance threshold.
11. Create selected feature set S from all retained features.
12. Split data into training and validation sets.
13. Train a baseline model using the original feature set X.
14. Train the same model using selected feature set S.
15. Compare validation performance using suitable metrics.

16. If performance is maintained or improved and dimensionality is reduced, accept S.
17. Otherwise, revise thresholds and repeat feature selection.
18. Return S.

END

Expected Result

The algorithm should reduce redundant and irrelevant attributes while retaining features that provide useful information for prediction. The validation step ensures that dimensionality reduction does not unnecessarily decrease model performance.

Question 3: Statistical Outlier Detection and Handling

Create a pseudocode algorithm to detect and handle outliers using statistical methods such as Z-score or IQR. The algorithm identifies extreme values, decides whether to remove or cap them, and verifies that the resulting dataset remains consistent and reliable.

Pseudocode

ALGORITHM Detect_And_Handle_Outliers

INPUT: Dataset D

OUTPUT: Cleaned dataset C

BEGIN

1. Load dataset D.
2. Identify all numeric columns.
3. For each numeric feature F:
 - a. Calculate mean μ and standard deviation σ .
 - b. Calculate Z-score for each value:
$$Z = (\text{value} - \mu) / \sigma$$
 - c. Mark values with $\text{ABS}(Z) > 3$ as potential outliers.
4. For each numeric feature F:
 - a. Calculate first quartile Q1.
 - b. Calculate third quartile Q3.
 - c. Calculate $\text{IQR} = Q3 - Q1$.
 - d. Calculate lower bound = $Q1 - 1.5 * \text{IQR}$.
 - e. Calculate upper bound = $Q3 + 1.5 * \text{IQR}$.
 - f. Mark values below the lower bound or above the upper bound as IQR-based outliers.
5. Combine the outlier information from the selected method.
6. Examine the proportion and distribution of outliers.
7. Select an action:
 - a. REMOVE outlier rows when they are invalid or erroneous.
 - b. CAP outliers at lower/upper limits when extreme values may represent valid observations.
 - c. KEEP outliers when domain analysis confirms they are genuine.
8. Apply the selected treatment.

9. Recalculate basic statistics after treatment.
10. Verify that no unintended missing or invalid values were created.
11. Compare the cleaned data distribution with the original data.
12. Store the resulting dataset as C.
13. Return C.

END

Consistency Check

After treatment, verify row counts, missing-value counts, minimum and maximum values, and summary statistics. This prevents accidental removal of valid observations or creation of inconsistent data.

Question 4: Dimensionality Reduction Using PCA

Write a pseudocode algorithm for Principal Component Analysis (PCA), including standardization, covariance-matrix computation, eigenvalue and eigenvector extraction, principal-component selection, efficient matrix operations, and validation of the reduced dimensions.

Pseudocode

ALGORITHM PCA_Dimensionality_Reduction

INPUT: Dataset D containing numeric features

OUTPUT: Reduced dataset R

BEGIN

1. Load dataset D.
2. Select the numeric feature matrix X.
3. Handle missing and invalid values.
4. Standardize every feature:

$$Z = (X - \text{mean}(X)) / \text{standard_deviation}(X)$$

5. Compute the covariance matrix:

$$\text{COV} = \text{COVARIANCE_MATRIX}(Z)$$

6. Compute eigenvalues and corresponding eigenvectors of COV using an efficient matrix decomposition method.
7. Pair each eigenvalue with its corresponding eigenvector.
8. Sort the eigenvalue-eigenvector pairs in descending order of eigenvalue magnitude.

9. Calculate explained variance ratio for each component.

10. Calculate cumulative explained variance.

11. Select the smallest number K of principal components that satisfies the required explained-variance threshold, such as 90% or 95%.

12. Construct the projection matrix W using the first K eigenvectors.

13. Project standardized data onto the selected components:

$$R = Z \times W$$

14. Verify that the number of rows in R equals the number of rows in the original dataset.

15. Verify that the number of columns in R equals K.
16. Confirm that cumulative explained variance meets the selected threshold.
17. Compare model performance using original and reduced data when PCA is being used for machine-learning purposes.
18. Return reduced dataset R.

END

Conclusion

The four algorithms provide systematic approaches for preparing and transforming machine-learning data. The first algorithm cleans and resamples time-series data, the second selects relevant and non-redundant features, the third identifies and treats statistical outliers, and the fourth reduces dimensionality using PCA. Together, these procedures improve data quality, reduce unnecessary complexity, and prepare datasets for reliable machine-learning analysis.